

Abstract Reasoning-Driven Prediction and Root Cause Discovery of Device Incidents in Core Internet Infrastructure

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Abstract. Telecommunication network incidents must be detected early and accompanied by plausible root causes identified from noisy, nonstationary observability streams, yet token-level pipelines are costly, brittle under log drift, and hard to explain. We propose an abstract-reasoning framework that parses raw log into log templates and treats each template as a concept token, enabling a compact concept-based Transformer, Abstract Reasoning LogTransformer (ART), to learn temporal dependencies via multitask next-template prediction and short-horizon incident prediction. This approach is partly inspired by reasoning paradigms found in large concept model and world model research in other fields. By abstracting logs into conceptual representations, our method reduces computational costs by $\mathcal{O}(\kappa)$ to $\mathcal{O}(\kappa^2)$ depending on the Transformer implementation, with $\kappa \approx 23$ denoting the average token count per log message. A learnable aggregation head assigns proxy incident-association scores to concept tokens, enabling ART to quantify the relevance of individual log concepts to the predicted incident while exposing their full underlying semantics, thereby facilitating prediction-time root cause analysis. In parallel, a dedicated template mining workflow enhances template induction throughput, making the framework suitable for carrier-scale datasets. On a real-world dataset from 1,213 Juniper routers and switches collected over 40 days with millions of log messages per hour and multiple incident modes, the proposed model achieves AUC ROC 0.96 and F1 score 0.78 under severe class imbalance. For reproducibility, implementation and experimental dataset ³ are publicly released online.

Keywords: Abstract reasoning · Concept model · Concept token · Root cause analysis · Device incidents · Core Internet infrastructure

1 Introduction

Network Operations Centers (NOCs) in telecom operators are responsible for maintaining the core IP and MPLS infrastructure that underpins broadband ac-

³ Code: GitHub, HuggingFace datasets: ML (LGBM) dataset & ART dataset

cess, mobile backhaul, content delivery, and enterprise connectivity. Even a single device incident can trigger routing reconvergence, protection switching, and traffic reoptimization, thereby degrading service-level objectives across multiple dependent services [9]. Early incident prediction is therefore operationally critical: it enables operators to prioritize verification, traffic steering, mitigation, or vendor escalation before customer-visible degradation emerges. In practice, operators also require associative evidence linking a warning to specific log events or device states, so that intervention can be strategically narrowed toward the most plausible failure mechanisms rather than downstream symptoms.

This problem remains difficult in real NOC environments. Observability log streams are massive, noisy, and highly nonstationary. Also, message patterns evolve with firmware updates, configuration changes, and emerging failure modes, while incident labels are relatively sparse. These properties make short-horizon incident forecasting challenging, and they make actionable incident reasoning even harder. A practical solution must therefore satisfy three requirements simultaneously: strong predictive performance, robustness to log-template drift, and interpretable evidence that can support operator action in real time.

Existing approaches do not fully meet these requirements. Traditional fault-management methods have long studied alarm correlation and fault identification in communication networks [4]. In operational ML pipelines, gradient-boosted decision trees remain strong approaches for heterogeneous telemetry [14], and supervised studies have explored network failure prediction in related domains [3]. However, such pipelines typically rely on sparse lexical features over raw or lightly normalized log, making them vulnerable to template drift and limiting the semantic interpretability of their alerts. Deep log models such as DeepLog and LogBERT capture richer sequential structure [11], but they are designed for anomaly detection rather than supervised short-horizon incident forecasting, and they provide limited support for transparent incident reasoning. More recently, large language models have also been explored as assistants for incident investigation and mitigation [1], yet direct token-level modeling of raw log remains computationally expensive and operationally unstable at carrier scale.

In this work, we introduce Abstract Reasoning LogTransformer (ART), an abstract associative reasoning framework for proactive device incident prediction and root cause identification in core Internet infrastructure. ART treats parsed log templates as concept tokens so as to reason over higher-level system signals. Built on a compact causal transformer architecture with multitask training, ART jointly learns next concept token prediction and supervised incident prediction. Its learnable linear aggregation head produces a proxy measure of the association of each concept token with incidents to support root cause analysis (RCA).

We evaluated ART on a production dataset from 1,213 Juniper routers and switches collected over 40 days, with log volume on the order of millions of messages per hour. ART achieves an AUC-ROC of 0.96 and an F1-score of 0.78 and the Spearman correlation coefficient between the proxy measure of incident association and empirical incident rates of concept tokens is $\rho = 0.52$. This enables more efficient root cause identification at prediction time. We further

show that concept-level template modeling substantially improves computational tractability relative to raw-token formulations, which reinforces its practicality for large-scale operational deployment.

2 Methods

2.1 Problems

The proposed methodology is formulated to address two fundamental tasks in large-scale network operations: short-horizon proactive device incident prediction and root cause analysis (RCA). To this end, we conduct experiments on high-volume log streams collected from Juniper network devices, where heterogeneous log events are aggregated into non-overlapping temporal windows per device.

Thus, we formulate incident prediction as a temporal learning task. Given recent observations of device-level activity, the model estimates the likelihood of an upcoming incident within a future horizon, capturing the need for proactive detection under strict time constraints in operational environments. To support both prediction and RCA, the input representation integrates sequential patterns from log messages with consolidated quantitative signals, enabling the model to reason multifactorially. This unified representation is critical for capturing both symptomatic patterns and underlying causal factors in complex network systems.

2.2 Well-established and effective machine learning approach

LightGBM as a top-notch model Prior work on operational incident prediction commonly employs tree-based ensembles, particularly gradient-boosted decision trees (GBDT), due to their effectiveness on heterogeneous and high-dimensional observability data [3,14]. This aligns with structured-learning benchmarks, where tree-based methods remain highly competitive and outperform deep learning models [10,14]. Thus, we adopt an efficient and scalable implementation of GBDT, Light Gradient-Boosting Machine (LightGBM) with standard configurations [14]. Its role in this work is to serve as a well-established non-sequential approach for benchmarking against our proposed ART model.

Feature engineering pipeline for log data

Lexical features via TF-IDF n -grams Raw log messages are parsed to extract structured fields (e.g., severity, facility), and the free-text content is normalized by removing high-entropy tokens such as identifiers and numeric fields. TF-IDF features are then computed over up to bigram vocabularies within each time window, producing a sparse representation of recurring log patterns.

Temporal dynamics beyond keywords To capture short-term deviations preceding incidents, we construct distribution-shift features over log volumes, stratified by severity (7 levels) and facility (23 categories). Specifically, ratios between the current and preceding windows are generated to detect relative bursts that are indicative of anomalous behavior across devices.

2.3 Concept-based modeling approach

Conceptual foundation Large Concept Models (LCMs) [2] advocate moving from token-level modeling to higher-level semantics, or *concepts*, for efficiency and abstraction. Furthermore, Causal Feature Learning (CFL) [6] suggests that fine-grained micro-level observations can often be aggregated into higher-level macro variables with more stable causal meaning. This perspective aligns with our use of log templates as abstractions of raw log messages, allowing ART to focus on incident-relevant patterns at a more semantically coherent level.

LogBERT [11] applies a BERT-based architecture [8] over log template sequences obtained via Drain parsing [12], treating each template as a discrete token. It applies self-supervised learning technique using masked log key prediction to model normal behavior and detect anomalies. However, LogBERT is designed for anomaly detection rather than supervised incident prediction, and its bidirectional masked objective does not explicitly model temporal causality between events. This limits its applicability for root cause analysis, where directional dependencies between log events and subsequent incidents are critical.

Abstract Reasoning LogTransformer We propose Abstract Reasoning LogTransformer (ART), a causal extension of LogBERT tailored for incident prediction and root cause analysis. ART introduces three key modifications. First, the masked language modeling objective is replaced with an auto-regressive next-token prediction objective, enforcing a unidirectional formulation that captures temporal dependencies between log events. Second, supervised incident prediction is also involved in the multi-task learning objectives, aligning sequence modeling with the downstream task. On top of that, we introduce a learnable linear aggregation head that assigns explicit token-level associations with incidents. Similarly to LogBERT, ART operates on log templates obtained via Drain parsing, treating them as concept tokens. This abstraction reduces noise from raw logs while enabling reasoning over higher-level system states.

Aggregation layer Rather than relying on conventional aggregation strategies, such as using the embedding vector of the [CLS] token or applying average pooling, which tend to dilute localized incident association signals by assigning uniform weight to all tokens, we employ a learnable linear aggregation layer, as illustrated in Fig. 1. This transformation assigns distinct, trainable weights to individual concept tokens in the input sequence, thereby producing a vector of concept token association measures. By enabling differential weighting across concept tokens, the aggregation layer explicitly captures the association structure underlying incident progression and identifies the most influential factors corresponding to specific event logs within the operational time window.

Multitask training objective ART is trained with a multitask objective [5] that combines supervised classification, an auxiliary next concept token prediction task, and an ℓ_1 sparsity regularization on the aggregation layer:

$$\mathcal{L}_{\text{total}} = \lambda_1 \mathcal{L}_{\text{incident}} + \lambda_2 \mathcal{L}_{\text{causal}} + \lambda_3 \mathcal{L}_{\ell_1}. \quad (1)$$

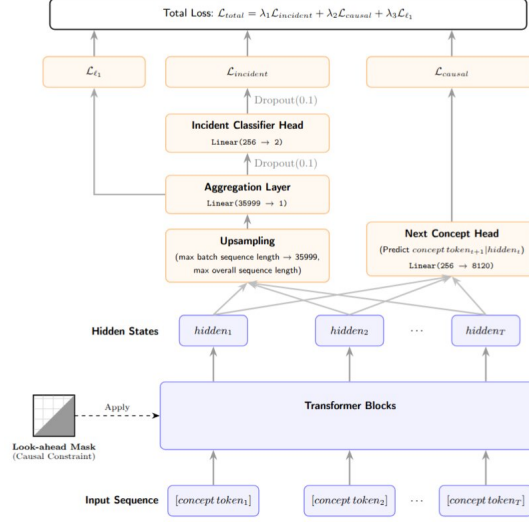


Fig. 1. Abstract Reasoning LogTransformer (ART) architecture

Here, $\mathcal{L}_{\text{incident}}$ is the classification loss for incident prediction, $\mathcal{L}_{\text{causal}}$ is an auto-regressive prediction loss with causal masking [16] that models temporal dependencies in log sequences, and $\mathcal{L}_{\ell_1}$ is an ℓ_1 regularization loss applied to aggregation layer weights to encourage the model to focus on a small subset of causally relevant log templates rather than distributing attention uniformly. In practice, we set $\lambda_1 > \lambda_2$ to prioritize the main classification objective.

Optimization strategy To reconcile variable-length log sequences with the fixed-length requirement of the aggregation layer, we adopt two further strategies.

1. **Bucket sampling.** Sequences are grouped by lengths into mini-batches [15], reducing padding overhead and improving computational efficiency.
2. **Post-encoder up-sampling.** Instead of padding at the input, sequence alignment is deferred until after the encoder. Variable-length representations produced by the transformer are up-sampled to a unified length, enabling consistent alignment between concept tokens and the aggregation weights and avoiding unnecessary computation on padded inputs.

3 Results

3.1 Experimental setup

Dataset We evaluate ART on a real-world NOC dataset collected from 58 Juniper network devices over a continuous 40-day period (November 21 to December 30, 2025). Although the operational infrastructure consists of 1,213 devices,

Table 1. Distribution of incident labels on Production

Label	Class/Incident type	Count	Percentage (%)	Relevant IPs
0	Non-Incident	550,311	93.27	58
1	All incidents	39,727	6.73	–
	Linecard down	–	5.15	22
	Traffic down	–	1.36	9
	Trap link down	–	0.20	31
	Log error	–	0.02	22

training and evaluation are restricted to the 58 devices that experienced at least one labeled incident, as including all devices would reduce the positive rate to below 0.4%, making supervised learning impractical due to the severe class imbalance. This restriction is further supported by operational statistics collected between January and April 2026, during which only 67 unique devices experienced incidents. The number of incident-prone devices remained consistently low, ranging from 14 to 35 devices per month (an average of 27 devices per month) or 32 to 54 devices over any two-month period, indicating that incidents on the remaining devices were persistently rare. Consequently, incident prediction for devices without historical incidents is of limited practical necessity. Furthermore, all baseline methods are evaluated on the same subset of 58 devices, ensuring a fair and unbiased comparison. As demonstrated in Section 3.2, although ART is trained exclusively on this subset, it still achieves a false positive rate (FPR) on the remaining devices that satisfies operational requirements. To prevent temporal leakage and reflect the intended production workflow, we adopt a strict chronological split: the first 20 days for training, followed by 10 days for validation, and the final 10 days for testing. Ground-truth labels cover five device failure modes: line card error, log error, trap link down, traffic down, and subscriber error. Tables 1 and 2 report the class distribution across incident types and the sequence length statistics with corresponding positive rates.

Table 2. Sequence length distribution and overall positive rate

Length range	Total count	Frequency (%)	Positive rate (%)
0–999	563,368	95.44	5.27
1000–1999	12,566	2.13	4.98
2000–2999	4,507	0.76	5.24
3000–3999	270	0.05	74.44
4000–4999	9,427	1.60	94.53
5000–9999	89	0.02	21.35
10000–19999	38	0.01	18.42
20000–29999	33	0.01	45.45
≥30000	13	0.00	0.00

Implementation details

ML pipeline Using the log-derived features from Section 2.2, including TF-IDF n-grams, distribution-shift features, and Levenshtein-ratio features, we apply mutual-information feature selection to retain the top 16,384 features. These are used to train a LightGBM model with validation-based hyperparameter tuning. In deployment, alerts are generated every 5 minutes using a ranking-based policy that selects a threshold or percentile based on available on-call capacity.

ART implementation For ART, raw log messages are first converted into log templates using Drain3 [12]. Aside from the standard Drain3 configuration, we apply domain-specific masking of high-entropy fields (e.g., timestamps, device identifiers, IP/MAC addresses, numeric identifiers, and URLs) to improve template quality for network logs. Our main customization is a length-sharded, multi-process template mining workflow. Instead of a single global Drain3 tree, messages are partitioned by token count after normalization, with an independent parser shard for each branch. Masking, delimiter normalization, and deduplication are performed in a separate parallel preprocessing stage before template induction. Compared with the default single-process pipeline, this design substantially improves throughput, reducing end-to-end template mining on 30 days of logs from about 20 days to under 20 hours.

To avoid leakage, the template inventory is induced using training logs only and then frozen for validation and test, accumulating a total vocabulary set of 8,120 concept tokens. For illustration, a representative sample before and after preprocessing and template induction is displayed as follows.

Raw: <pre><13>Dec 1 12:27:22 VLG-AggPE-01-02 ppman[8452]: %USER-5: ... Lacp pdu change, Send to RE iflIdx: 1052</pre> Parsed: <pre>13 TIME DEVICE ppman PID %USER-5 ... Lacp pdu change, Send to RE iflIdx NUM</pre>
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ART instantiates a compact causal Transformer encoder (hidden size 256, one Transformer block; 4.73M parameters). As described in Section 2, sequence bucketing and post-encoder up-sampling follow the method design, and training optimizes the multitask objective in Eq. 1, with $\lambda_1 = 1$, $\lambda_2 = 0.1$ and $\lambda_3 = 10^{-4}$.

3.2 Incident prediction

At the optimal thresholds to balance precision and recall (aligning with operational resource constraints), both the ML and the proposed ART model demonstrate strong predictive performance. As summarized in Table 3, the LightGBM achieves a precision of 0.91 and a recall of 0.67 ($F1 = 0.77$), while the ART model yields comparable results with a precision of 0.93 and a recall of 0.68

Table 3. Performance Comparison between LightGBM and ART

Model	AUC	Precision	Recall	F1-Score
LightGBM	0.96	0.91	0.67	0.77
ART	0.96	0.93	0.68	0.78

($F1 = 0.78$). Both achieves an AUC-ROC score of 0.96, demonstrating their practical effectiveness for early-warning triage under strict resource constraints.

Specifically, the evaluation was conducted on a 10-day test set comprising 159,511 device-windows (samples), including 7,716 positive samples (a 4.84% positive rate), which represent the complete set of observed incidents. The reconstructed confusion matrix yields 5,248 true positives, 421 false positives, and 151,795 true negatives. The 421 false positives generated over the 10-day period represent only a 5.46% increase relative to the number of actual incidents. This corresponds to approximately 42 false alerts per day, 1.75 false alerts per hour, or fewer than one false alert per device every 1.4 days. Such a false alarm rate is well within operational requirements and aligns with the primary objective of network operations, namely maximizing incident detection coverage to minimize the risk of cascading failures in core network devices that could potentially affect millions of customers. Consequently, the proposed approach enables earlier incident detection and contributes to reducing the Mean Time to Identify.

Furthermore, in the production deployment of ART, the set of incident-prone devices is updated on a daily basis as new incidents are observed. To evaluate the model’s generalization capability to newly emerging incident-prone devices, we conducted an additional experiment using the 67 devices that experienced incidents between January and April 2026. Specifically, we assessed the false positive rate (FPR) over a 10-day testing period on these devices. Under this setting, ART maintained 99.7% specificity, producing 164,143 true negatives and 430 false positives across 164,573 device-windows, corresponding to an out-of-sample FPR of 0.26%. In operational terms, this corresponds to at most two false alerts per hour. In practice, both the model and the set of monitored incident-prone devices are updated daily as new incidents are observed. Consequently, the number of false alerts is expected to be even lower than that observed under this static evaluation setting. Therefore, the resulting alert volume remains well within the operational handling capacity of the 21-member NOC team.

Nevertheless, interpretability remains a key trade-off. Traditional ML models can achieve competitive accuracy and are easier to deploy, but their TF-IDF tokenization limits reasoning to micro-level lexical patterns that often lack coherent semantics when decoupled from their original log context. This bag-of-tokens representation makes it difficult for operators to understand fault mechanisms or log event sequences. In contrast, ART utilizes concept-level representations acting as macrovariables [6], where each token represents a single, complete log template possessing its full underlying semantics. This abstraction provides more transparent reasoning and better supports root cause analysis. Unlike individual

lexical tokens that may recur across disparate event types, thereby “mixing” statistical correlations and confounding the model’s reasoning, each concept token in ART corresponds to a unique, atomic log event. This mapping ensures that incident association weights, derived via the post-encoder up-sampling technique, are explicitly and cleanly assigned to specific log events, as further examined in Subsection 3.3.

3.3 Association analysis and root cause identification

To support incident root cause investigation, we estimate the rank $r(t)$ of $P(Y = 1 | t)$, where t denotes a concept token appearing in an input sequence s . The objective is to prioritize log patterns that help experts narrow plausible root causes. In ART, the rank of the learned weights $\mathbf{w}$ from the linear aggregation layer, as specified in paragraph 2.3, is used as a proxy for $r(t)$. We evaluate this approximation by computing the Spearman rank correlation between the weights for concept token t and the empirical conditional probabilities $P(Y = 1 | t)$.

Average concept token association with incidents Due to the post-encoder up-sampling mechanism, the association weight of a concept token t of s is distributed across multiple indices of $\mathbf{w} = (w_i)$ without being mixed between different concept tokens. The set of all such indices is denoted as $\mathcal{I}_{s,t}$. We define $\hat{w}(s, t)$ as a proxy measure of the association of concept token t within the sequence s on incidents and the average association over the dataset $\hat{w}(t)$ calculated as:

$$\hat{w}(s, t) = \sum_{i \in \mathcal{I}_{s,t}} w_i \quad \text{and} \quad \hat{w}(t) = \frac{1}{N_t} \sum_{s=1}^{N_t} \hat{w}(s, t) \quad (2)$$

where N_t is the total sequences in dataset containing t . We evaluate the effectiveness of $\hat{w}(t)$ using the Spearman rank correlation coefficient against the observed incident rate $P(Y = 1 | t)$.

Experimental results We evaluated $\hat{w}(t)$ on 400 concept tokens followed by incidents out of a vocabulary of 8,120 concept tokens. Evaluating the empirical incident rate $\hat{P}(Y = 1 | t)$ at the individual concept token level can be highly noisy. To reduce this noise and improve estimation stability, we sort the concept tokens in ascending order of $\hat{P}(Y = 1 | t)$ and partition them into 20 equally populated bins. For each bin, we compute the average of both the empirical incident rates and the estimated association measures $\hat{w}(t)$ to serve as representative values.

Fig. 2 visualizes ART’s RCA capability with a rank-rank plot. The Spearman correlation of $\rho = 0.52$ shows a clear positive relationship: concept tokens with higher rankings in estimated association scores tend to have higher rankings in incident likelihoods. This suggests that ART can rank plausible root causes at forecasting time.

Finally, operational rationality is validated in Table 4. Log templates exhibiting high incident rates and large $\hat{w}(t)$ align perfectly with early warning signals

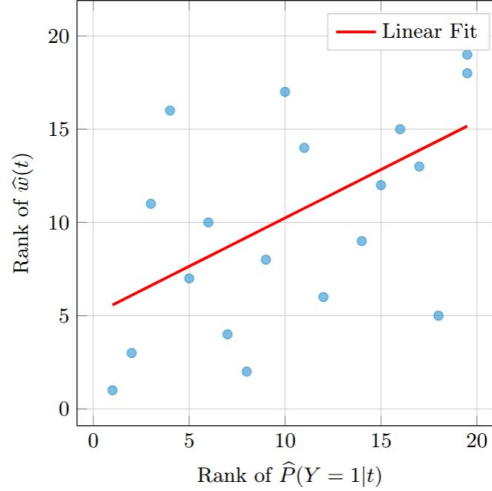


Fig. 2. The scatter plot shows the relationship between the ranks of incident rates and estimated association measures across the 20 concept-token bins, with a Spearman correlation of $\rho = 0.52$.

monitored by NOC engineers on Juniper devices, such as forwarding plane stress or control plane protocol errors.

3.4 Computational and parameter efficiency

A key practical advantage of concept-level templates is their shorter effective sequence length. Let L be the concept-token sequence length used by ART. As shown in Table 2, L can exceed 30,000 in the long tail. A raw-token formulation would expand each concept token into κ tokens, giving $L_{\text{token}} = \kappa L$. In our data, $\kappa \approx 23$ on average, making the required context length far beyond what standard transformers can handle efficiently in training or inference.

The gap is further amplified by the sequence-length dependence of self-attention. Standard transformer attention scales as $O(n^2)$ [17], while efficient variants such as sparse transformer, reformer, and linformer reduce this cost through sparse or approximate attention mechanisms as shown in Table 5 [7,13,18].

When moving from concept tokens to raw tokens, effective sequence length increases from L to $L_{\text{token}} = \kappa L$. Under standard self-attention, this implies an approximate κ^2 increase in attention cost. With $\kappa \approx 23$ in our log data, the raw token formulation would require $\approx 529\times$ more attention computation. Although efficient attention variants reduce this scaling, the raw token formulation remains substantially more expensive for long log streams. This estimate also does not account for the additional parameter and training overhead typically associated with finer-grained modeling, including larger embedding and FFN layers with increased hidden size (e.g., $d: 256 \rightarrow 512$). In contrast, ART remains compact

Table 4. Representative Juniper NOC log templates with high incident rates.

Full log template string	$\hat{P}(Y=1 t)$	$\hat{w}(t)$
... mib2d ... %DAEMON-3-MIB2D_COUNTER_DECREASING pfes_stats_delta counter PFES_TRAFFIC_FABRIC_DROPS decreasing ... new <*> old <*>	1.0	0.32
... mib2d ... %DAEMON-3-MIB2D_COUNTER_DECREASING pfes_stats_delta counter PFES_TRAFFIC_INPUT decreasing ... new <*> old <*>	1.0	0.26
... rpd ... %DAEMON-3-JTASK_SEND_RECV_ERROR Send msg call failed for task PIM Recv6 ... Device not configured	0.5	0.17

Table 5. Self-attention complexity comparison

Model Architecture	Complexity per Layer	Sequential Operation
Transformer [17]	$O(n^2)$	$O(1)$
Sparse Transformer [7]	$O(n\sqrt{n})$	$O(1)$
Reformer [13]	$O(n \log n)$	$O(\log n)$
Linformer [18]	$O(n)$	$O(1)$

(4.73M parameters, hidden size 256, one Transformer block) while achieving competitive performance on this domain-specific task (Table 3).

4 Discussion

The experimental results demonstrate that ART successfully addresses the twin challenges of proactive incident prediction and root cause analysis in core network infrastructure. With an AUC-ROC of 0.96 and an F1-score of 0.78 shown in Table 3, our approach delivers competitive predictive performance while offering interpretable insights into incident precursors.

The main contribution of ART therefore lies not in a dramatic gain in aggregate classification metrics, but in combining competitive early-warning performance with a more informative and efficient representation of incident precursors. By operating on parsed log templates rather than raw log tokens, ART reduces sensitivity to high-entropy lexical variation and potential template drift, allowing the model to focus on more stable system-state patterns. This abstraction also substantially reduces computational complexity proportionally to the effective sequence length, rendering it far more efficient on large-scale log streams than raw token formulations. Thus, ART improves both explainability and deployment practicality as its concept token–incident associations are more actionable for NOC workflows and can help reduce Mean Time to Identify.

Despite these promising results, several limitations remain. First, the approach depends on accurate template induction: parser errors or semantically ambiguous logs can affect the learned representation. Second, the concept token association measures reflect associations learned under the observed distribution

rather than fully identified causal effects. Stronger causal claims would require intervention-based validation, which is rarely feasible in production networks.

Overall, these results suggest that concept-level causal sequence modeling is a promising direction for proactive network operations. ART remains competitive with top-performing models such as LightGBM, while also contributing interpretable root-cause signals and improved computational efficiency on long log sequences, two advantages that conventional tabular models do not naturally provide. A natural next step is to integrate ART with LLM-based decision-support tools to contextualize concept token associations, generate structured incident summaries, and support human-in-the-loop remediation planning.

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